Translational Science

Cancer Research

Gaussian Mixture Models for Probabilistic Classification of Breast Cancer

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Abstract

In the era of omics-driven research, it remains a common dilemma to stratify individual patients based on the molecular characteristics of their tumors. To improve molecular stratification of patients with breast cancer, we developed the Gaussian mixture model (GMM)–based classifier. This probabilistic classifier was built on mRNA expression data from more than 300 clinical samples of breast cancer and healthy tissue and was validated on datasets of *ESR1*, *PGR*, and *ERBB2*, which encode standard clinical markers and therapeutic targets. To demonstrate how a GMM approach could be exploited for multiclass classification using data from a candidate marker, we analyzed the insulin-like growth factor I receptor (IGF1R), a promising target, but a marker of uncertain importance in breast cancer. The GMM defined subclasses with downregulated (40%), unchanged (39%), upregulated (19%), and overexpressed (2%) *IGF1R* levels; inter- and intrapatient analyses of *IGF1R* transcript and protein levels supported these predictions. Overexpressed IGF1R was observed in a small percentage of tumors. Samples with unchanged and upregulated IGF1R were differentiated tumors, and downregulation of IGF1R correlated with poorly differentiated, high-risk hormone receptor–negative and HER2-positive tumors. A similar correlation was found in the independent cohort of carcinoma *in situ*, suggesting that loss or low expression of IGF1R is a marker of aggressiveness in subsets of preinvasive and invasive breast cancer. These results demonstrate the importance of probabilistic modeling that delves deeper into molecular data and aims to improve diagnostic classification, prognostic assessment, and treatment selection.

Significance: A GMM classifier demonstrates potential use for clinical validation of markers and determination of target populations, particularly when availability of specimens for marker development is low.

Introduction

Breast cancer is a group of heterogeneous neoplasms that affects millions of women worldwide (1). While clustering of breast cancers into the molecular subtypes provided a new approach for estimation benefits and risks of current treatments (2), markers capable of influencing new diagnostic and treatment decisions are very slow in coming (3). Clinically, breast tumors are classified on the basis of morphology and status of HER2 and steroid hormone receptors (HR), such as estrogen receptor alpha (ERα) and progesterone receptor (PR), assessed as positive and negative mainly by IHC. Receptor status determines the suitability of using targeted treatments, such as trastuzumab and tamoxifen for HER2-positive ($HER2^+$) and ER-positive (ER^+) tumors, respectively (4, 5). Due to technical variabilities and inability to generate quantitative data by IHC, substantial efforts have been dedicated to the validating mRNA-based determination of the protein status to provide further basis for treatment decision making (6–8). The majority of gene expression studies have used a single cutoff point, either a published value (9), a certain sample quantile like the median (10), percentile (11), tertile (8), or an "optimized" cutoff, the value that correlates with binary clinical outcome and survival data (6, 7, 12). Although data dichotomization has gained widespread popularity in breast cancer research, there is uncertainty about the actual existence of a single cutoff for a given gene expression data and the potential analytic errors that may occur during molecular classification of cancer based on dichotomization (13, 14).

In the effort to better leverage a molecular variable's continuum in cancer classification, we present the probabilistic Gaussian mixture model (GMM)-based analytical method, which considers the multiple cutoff points with confidence (MCPC) and class discovery. MCPC discovery refers to computing the classification thresholds for a variable with given probabilities of being a member of each possible group. Class discovery refers to defining previously unrecognized groups. GMM is a powerful parametric tool that takes into account the multimodal distribution of numerical variables across all the measurements. Although trained by a particular finite dataset, this generative model is able to generate data. Through output of a probability of the variable, this model handles missing data, extracts much more information from small

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datasets, and thus may save time and money, by reducing the costs incurred by misclassification. As an efficient tool for complex data analysis, GMMs have been successfully applied in the fields of signal, image, and video processing (15, 16). In biology, GMMs have been used in protein mass spectra analysis (17), atomic modeling (18), quantitative genetics (19), and gene clustering (20).

In this article, we expand the scope of GMM application in biomedicine by developing a classifier with which to assign individual cancers into their respective subclasses, considering natural complexity in the expression of a molecular variable. We compared the classifier's effectiveness with binary algorithms on our datasets from the human genes' mRNA encoding HR and HER2. The classification of primary breast cancer based on GMM was consistent with the classification on standard clinical markers, yet GMM provided additional layers of information, including cutoffs with confidence for better determination of cancer suitability for targeted therapy. Furthermore, we explored the classifier's capability to delve deeper into development of markers that have not been exploited clinically due to incorrect assumptions or limited observational data. To this end, we investigated the insulin-like growth factor I receptor (IGF1R), a marker with a controversial diagnostic and prognostic value in breast cancer (21, 22) and a promising target for anticancer drugs that failed in more than 20 clinical trials performed in unstratified populations since 2008 (23). The probabilistic approach in our study allowed us to identify a subclass of IGF1R–overexpressing cancer, most likely to benefit from IGF1R–blocking therapy, and to emphasize the potential importance of downregulated IGF1R in the diagnostic and prognostic assessment of invasive and preinvasive breast tumors.

Materials and Methods

Human cell lines and breast tissues

The breast cancer cell lines SK-BR-3, BT-474, T-47D, MDA-MB-231, and MCF7, expressing various levels of HER2, ER, PR, and IGF1R, were purchased from the ATCC and cultured accordingly. MCF7–derived clones expressing high levels of IGF1R were from our laboratory (24). *Mycoplasma*-free cells, as confirmed by MycoAlert Assay (Cambrex), were used in the experiments. Formalin-fixed, paraffin-embedded (FFPE) breast tissues from surgical excisions and diagnostic core needle biopsies were used. Studies were conducted in accordance with recognized ethical guidelines.

An approval from the University of Pennsylvania Institutional Review Board committee with a waiver of written informed consent was used for analysis of patients' tissue and records. The study was performed with 394 samples, including 49 breast reduction mammoplasties from women with no history of breast cancer, 97 carcinomas *in situ* (CIS) on core needle biopsy, 150 invasive breast carcinomas (IBC), and 98 histologically normal matched tissues from surgical excisions in patients with breast cancer. Clinical and pathologic variables recorded from pathology reports and included for analysis were age, ER, PR, and HER2 status in all tumors; nuclear grade in CIS; and tumor size, histologic grade, and nodule status in IBC. Histologic grade was assessed according to a 3-tier (low, intermediate, and high) system based on the evaluation of three morphologic parameters, namely the percentage of the tumor arranged in glands and tubular structures, the degree of nuclear pleomorphism, and the mitotic rate. In Supplementary Table S1, the characteristics of the study population and clinical characteristics of tumor samples in which gene expression was measured are presented in accordance with the Reporting Recommendations for Tumor Marker Prognostic Studies recommendations (25).

Direct mRNA profiling

For mRNA profiling, we ran QuantiGene Plex 2.0 (QG2) Assay (GenoSpectra/Panomics/Affymetrix/eBioscience/Thermo Fisher Scientific) and read on FlexMap 3D (Luminex/Merck Millipore) according to the manufacturer's protocol. We analyzed measurements of specific probes, designed for use with the QG2 assay kits for quantitation of multiple target-specific RNAs directly in lysates from FFPE tissue and cell lines. Our human Plex Set 12988 included 16 target-specific and two housekeeping gene probes. Here, we analyzed mRNA for *ERBB2* (probe set region 1,203–1,621); *ESR1* (probe set region 5,671–6,292); *PGR* (probe set region 2,609–3,194); and *IGF1R* (probe set region 1,406–1,866), encoding HER2, ERα, PR, and IGF1R, respectively. Among the two housekeeping genes *PPIB* (peptidylprolyl isomerase B; probe set region 100–610) and *GUSB* (glucuronidase beta; probe set region 780–129), *PPIB* expressed consistently, whereas *GUSB* showed good expression in cultured cells, but relatively low expression level in FFPE tissue. Signal is reported as median fluorescence intensity and is proportional to the number of target mRNA molecules present in the sample. After subtracting background values from each measurement, the relative target gene activity in FFPE samples was calculated as a ratio: [(target gene expression − background)/(PPIB gene expression − background)] × 100. Samples with mRNA expression levels below the limit of detection were excluded (according to the recommendations from the assay provider). Test tissue sample lysates, obtained from five tissue curls of 10 μm each, and an array of internal negative and positive samples to control for technical variability was included in each run (Supplementary Table S2).

IHC

The status of steroid HR and HER2, determined by the FDA-approved methods, was obtained from surgical pathology reports. HER2 status in CIS was assessed by P.J. Zhang on the basis of the FDA-approved method for IBC. IGF1R staining was performed on a Leica Bond-IIITM instrument using the Bond Polymer Refine Detection System (Leica Microsystems AR9800) with positive and negative breast tissue control included in every run. Briefly, slides were dewaxed by baking, deparaffinized in Xylene, and followed by hydration with washes in 95%, 80%, and 70% ethanol and water. Heat-induced epitope retrieval with ER1 or ER2 solution (Leica Microsystems AR9640) at 99°C, then endogenous peroxidases block for 5 minutes in 1% hydrogen peroxide were followed by incubation with a primary mouse antibody raised against amino acids 741–800 of the IGF1R beta subunit of human origin (F-1, sc-390130, Santa Cruz Biotechnology, Inc.) at 1:200 dilution at room temperature. After primary antibody and three washes with bond buffer, slides were incubated for 8 minutes with a secondary anti-mouse IgG antibody, followed by washes and incubation with an anti-rabbit horseradish peroxide polymer for 8 minutes [note: an anti-IGF1R beta mAb is more specific and eliminates background of a commonly used anti-IGF1R beta (c-20; SC-713) polyclonal antibody; ref. 26].

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Protein profiling

The algorithm for measurement of the relative protein expression in IHC-stained tissue using ImageJ v.1.37v. software (NIH, Bethesda, MD) was validated (26). Using this algorithm, we developed a procedural computer-aided image (COMPAi) analysis software built on the image processing libraries of MATLAB v.15b (The MathWorks, Inc.). Measurements by ImageJ and COMPAi were compared with the Bland–Altman plot showing the acceptable average of the differences (bias = −0.049; SD = 0.1038) as illustrated in Supplementary Fig. S1. COMPAi maps normal epithelia, tumors, and areas of stroma, using the tissue segmentation function. It delivers a measure of the aggregate protein expression in the cell membrane, cytoplasm, and nuclei relative to stroma (i.e., background). The output variable is a relative pixel intensity computed on a continuous scale as a ratio: mean pixel intensity background/mean pixel intensity in region of interest (ROI), where ROI is automatically mapped. Manual inspection of the segmentations was performed to insure quality and accuracy. The mean value was computed from of at least five measurements per sample.

Statistical analysis and probabilistic modeling

The degree of agreement (concordance) between two methods measuring gene expression products, mRNA by QG2 assay, and protein by IHC, were assessed using two models: a receiver operating characteristic (ROC) and a fit of a mixture bimodal distribution analysis using Cutoff Finder, a comprehensive web application specialized to dichotomization (27). To choose a cutoff value in the ROC curve analysis that separates "negative" from "positive" test results, we calculated AUC with measurements sensitivity and specificity, as well as cutoff points. The latter were determined by minimizing the Euclidean distance between the ROC curve and left top edge of the diagram. To calculate ORs, logistic regression was executed using the function glm from R package stats (22). The strength of the relation between two variables was assessed by computing Pearson correlation coefficient (r) and Spearman rank correlation coefficient (*rho*). For significance test of each pairwise comparison of two groups we used the unequal variances t test. A two-sided significance level of alpha = 0.05 was used for all tests of significance. All statistical analyses were carried out in StatPlus Pro 6.3.0.5 (AnalystSoft, Inc.) and GraphPad Prizm 7.0 (GraphPad Software, LLC,).

The modeling of each single-gene expression data in the population of patients with breast cancer was based on finite GMM. The following analytical expressions were used to fit a series of Gaussians into histograms of data:

$$P(x) = \sum_i p_i(x) = \sum_i a_i e^{-(x-b_i)^2/2c_i^2},$$

where $p_i(x)$ is the posterior probability density for observing the gene expression level at x in the group i; a_i, b_i, and c_i are the weight, mean value, and SD for each single Gaussian component, respectively. The procedure of nonlinear least squares curve fitting was implemented to calculate a, b, and c in the analytical expressions. To improve the model and avoid overfitting, a choice of histogram bin widths and degrees of freedom was based on the adjusted R^2 value. The overall analytical expression for each gene was normalized to 1 to make it a probability density function (PDF):

$\int_{-\infty}^{\infty} P(x)dx = 1$, where $P(x)$ as above.

The proportion of patients in each group, that is, the prior probability for observing a patient in group i, was given by integral under the PDF component:

$P_i\ (x) = \int_{-\infty}^{\infty} p_i(x)dx$, where $p_i(x)$ as above.

For group classification of gene expression data, we used Bayes' theorem to define a confidence as the probability for an observation to belong to the Gaussian component i:

$$C_i(x) = \frac{p_i(x)dx/P_i}{\sum_j p_j(x)dx} P_i = \frac{p_i}{\sum_j p_j},$$

where $p_i(x)dx/\ P_i$ is the probability for observing the gene expression level at x in group i; $\sum_j p_j(x)dx$ is the probability for observing the gene expression level at x; and P_i is the probability for observing group i. The integration of PDF and optimization of curve fitting was done using the MATLAB v.17a Curve Fitting Toolbox (The MathWorks, Inc.).

Data and materials availability

All data needed to evaluate the conclusions in the article are present in the article or the Supplementary Information. COMPAi is available as a standalone application to qualified users.

Results

Data collection using QG2 assay and archival FFPE normal and cancer tissue

In this article, we used a highly reliable and validated multiplex branched-chain DNA(b-DNA) QG2 assay for parallel gene expression profiling in FFPE tissue (28, 29). This assay does not require RNA extraction, cDNA synthesis, or PCR amplification; thus, it minimizes the readout noise. Furthermore, this assay is suitable for analysis of archival FFPE tissue samples, in which procedural degradation of mRNA occurs naturally. We directly and concurrently measured classic and candidate markers of cancer progression mRNA expression in a cohort of women with no history of breast cancer and in a cohort of patients who were diagnosed with CIS and IBC. Sampling was done at random over 3 years of research with 394 samples analyzed in total. After excluding data from the low-abundance housekeeping gene–expressing samples, the relative activity of target genes was calculated in 328 FFPE samples. The success rate of gene expression profiling was 69.4% (34/49) in breast reductions, 77.3% (75/97) in CIS, 94.7% (142/150) in IBC, and 78.6% (77/98) in histologically normal samples matched to IBC. Thus, our study shows that direct measurement of gene's mRNA in standard FFPE surgical excisions and diagnostic core biopsies could be attained at an overall success rate of 83.2%.

Low rather than high IGF1R associates with and may predict aggressive breast cancer

In this article, we focused on the QG2 assay data for *ESR1*, *PGR*, *ERBB2*, and *IGF1R*, all of which are known to be important for the pathogenesis of breast cancer. We assessed the strength of the associations between the relative levels of each gene mRNA and clinicopathologic parameters (patient age at diagnosis, tumor size, lymph node status, histologic grade in IBC, and nuclear grade in CIS). The expression of *ESR1*, but not other analyzed target genes, correlated with age of patients at the time of mammoplasty ($r = 0.314$; $P = 0.07$), and surgery for CIS ($r = 0.364$; $P = 0.001$), and IBC ($r = 0.299$; $P = 0.0003$). This indicates that age is a critical factor to take into account to avoid the overestimation of ER positivity in breast cancer. As summarized in Table 1, there was no significant correlation found between tumor size, lymph node

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Table 1. Associations between *ERBB2*, *ESR1*, *PGR*, and *IGF1R* expression and clinicopathologic parameters in patients diagnosed with CIS and IBC

Gene	Parameter	n^a	*rho*	P^b
	Nuclear grade CIS			
ERBB2		75	0.11818	0.31258
ESR1		75	−0.19887	0.08717
PGR		75	−0.18304	0.11599
IGF1R		75	−0.2990	**0.00917**
	Histologic grade score IBC			
ERBB2		132	−0.00865	0.92154
ESR1		132	−0.3716	**0.00001**
PGR		132	−0.3804	**6.815E−6**
IGF1R		132	−0.4686	**1.457E−8**
	Glandular/tubular differentiation IBC			
ERBB2		100	−0.06703	0.5076
ESR1		100	−0.26744	**0.00715**
PGR		100	−0.23683	**0.01768**
IGF1R		100	−0.32709	**0.00089**
	Nuclear pleomorphism IBC			
ERBB2		100	0.07678	0.44767
ESR1		100	−0.32471	**0.00098**
PGR		100	−0.32674	**0.00091**
IGF1R		100	−0.37704	**0.00011**
	Mitotic count IBC			
ERBB2		100	−0.09146	0.36545
ESR1		100	−0.4084	**0.00002**
PGR		100	−0.38568	**0.00007**
IGF1R		100	−0.34705	**0.0004**
	Tumor size IBC			
ERBB2		141	0.03274	0.69992
ESR1		141	0.0817	0.33551
PGR		141	0.01964	0.81716
IGF1R		141	−0.07595	0.37074
	Nodule status IBC			
ERBB2		130	0.04387	0.62017
ESR1		130	0.08304	0.34759
PGR		130	0.04771	0.58989
IGF1R		130	−0.11156	0.20638

[a]*n*, number of samples in group.
[b]An association found to be statistically significant is shown in bold.

status, and expression of the four tested genes. However, there was a significant correlation between lower expression of *IGF1R* and higher nuclear grade in cancer *in situ* (rho = −0.30; $P = 0.009$). In addition, higher histologic grade in primary IBC was associated significantly with downregulation of three genes: *IGF1R* (*rho* = −0.47; $P = 1.458E\text{-}8$), *PGR* (*rho* = −0.38; $P = 6.81559E\text{-}6$), and *ESR1* (*rho* = −0.37; $P = 0.00001$). Furthermore, *IGF1R* showed a significant inverse correlation with every factor that pathologists take into consideration to grade aggressiveness of IBC, that is, nuclear pleomorphism (*rho* = −0.4; $P = 0.0001$), differentiation (*rho* = −0.3; $P = 0.0009$), and mitotic count (*rho* = −0.4; $P = 0.0004$). As summarized in Table 2, the ROC curve analysis revealed a good ability of the *IGF1R*, *ESR1*, and *PGR* mRNA measurements to predict the more aggressive cancer, with *IGF1R* being the best predictor of high nuclear grade *in situ* [AUC = 0.70; 95% confidence interval (CI), 0.55–0.85; $P = 0.004$) and high histologic grade IBC (AUC = 0.75; 95% CI, 0.66–0.84; $P < 0.0001$). These results raise the possibility that IGF1R is a marker of differentiation and that lack of its expression may be involved in the pathogenesis of *in situ* and invasive breast cancer.

GMM curve fitting and cancer classification using *ERBB2*, *ESR1*, *PGR*, and *IGF1R* mRNA

The significant association between the histologic factors of cancer aggressiveness and low, rather than high, *IGF1R* expression brought into question the rates of occurrence of low and high expression of *IGF1R* in the human breast cancer. To address this outstanding question, the nontrivial question of how to determine which levels of mRNA are "normal" and "abnormal" must be addressed. The majority of studies on *ERS1*, *PGR*, and *ERBB2* expression used ROC curve analysis and a two-component Gaussian mixture distribution to define the optimum single cutoff for mRNA data with which to assign a cancer sample to one of two groups (6, 27, 30). Although in these studies the absolute value of a cutoff for mRNA varied between gene profiling platforms, for example, DNA microarray or reverse transcription quantitative PCR, the binary classification for mRNA data was common among the studies. Using this statistical concept first, we evaluated how

Table 2. ROC curve analysis evaluation of *ERBB2*, *ESR1*, *PGR*, and *IGF1R* as potential biomarkers of high nuclear grade in CIS and high histologic grade in IBC

Breast cancer	Gene	AUC	95% CI	*P*
In situ				
	ERBB2	0.5699	0.424–0.7158	0.3158
	ESR1	0.5	0.364–0.636	>0.9999
	PGR	0.6615	0.5338–0.7892	**0.0204**
	IGF1R	0.7003	0.5549–0.8458	**0.0040**
Invasive				
	ERBB2	0.5108	0.408–0.6135	0.8403
	ESR1	0.6811	0.5853–0.7768	**0.0007**
	PGR	0.6834	0.5835–0.7833	**0.0006**
	IGF1R	0.7515	0.6631–0.8400	**<0.0001**

NOTE: Models found to be statistically significant are shown in bold.

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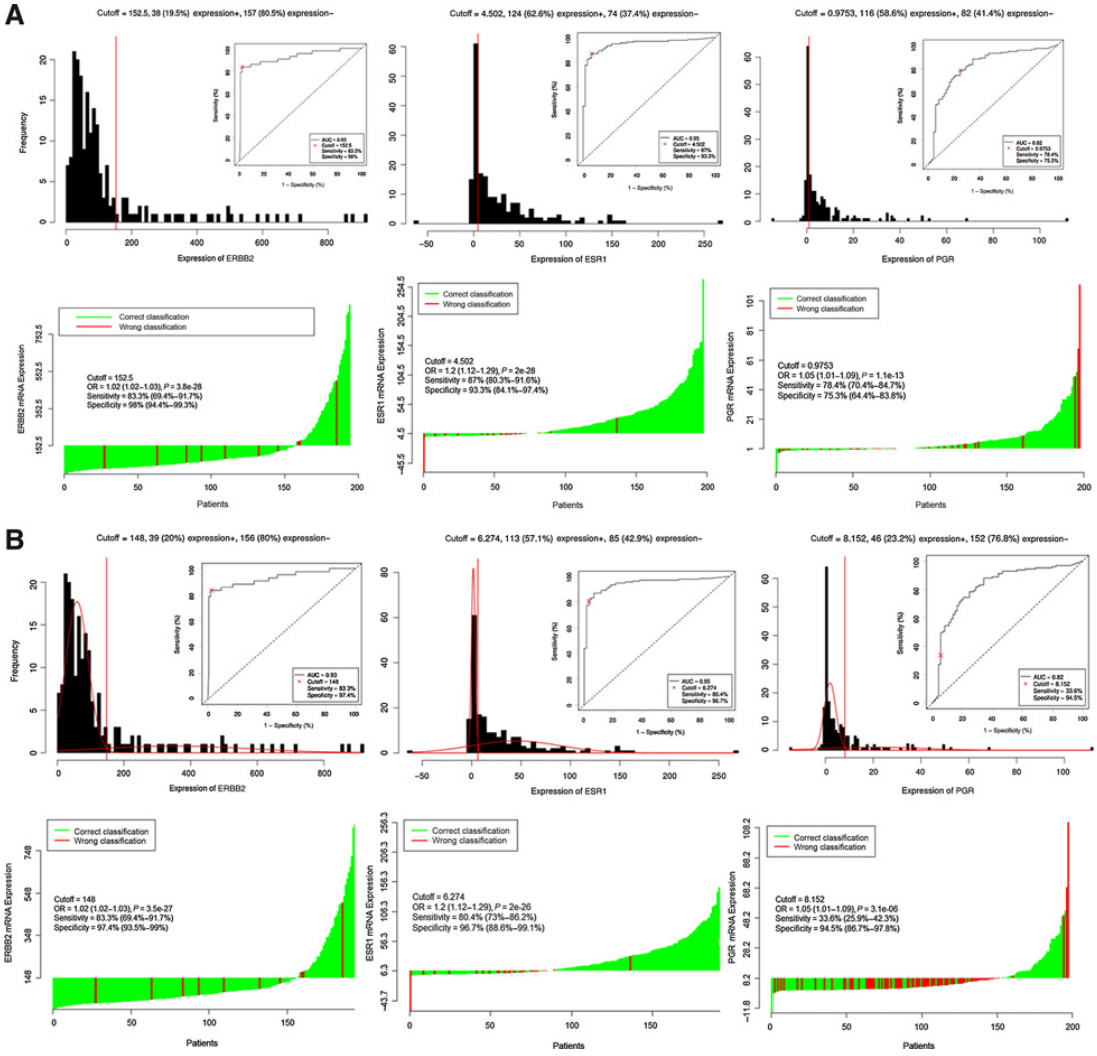

Figure 1.

Concordance between QG2 assay data and pathology reports on HER2, ER, PR status. **A,** Optimal cutoffs for each gene calculated on the basis of ROC curve analysis using AUC, sensitivity, and specificity. **B,** Optimal cutoffs for each gene calculated on the basis of the dichotomization (positive vs. negative on IHC); a mixture model of two Gaussians distributions fitted to each of the histograms (red curves). Histograms for *ERBB2* ($n = 195$), *ESR1* ($n = 198$), *PGR* ($n = 198$) expression in breast carcinoma with available status for HER2, ER, PR (top rows). Waterfall plots for the fixed cutoffs (bottom row). Red vertical lines are the optimal cutoffs derived in each model. All figures were automatically generated by Cutoff Finder.

well the QG2 assay measurements of *ERBB2*, *ESR1*, and *PGR* mRNA predicted IHC outcomes (Fig. 1A and B). In a ROC model, the *ERBB2* mRNA data and IHC scoring for HER2 had a concordance of 93%, with a positive agreement of 83% and a negative agreement of 98%. The optimal cutoffs for *ERBB2* data determined by two methods were rather similar: 152 and 148. For *ESR1* mRNA and IHC-based ER scoring, concordance was as high as 95%, with a positive agreement of 87% and a negative agreement of 93%. The cutoff points defined for *ESR1* were comparable at 4.5 and 6.3, respectively. Notably, *PGR* mRNA and a binary status of PR were less concordant (82%), with a positive agreement of 78% and a negative agreement of 75% in the ROC model. Furthermore, the ROC and bimodal fit model determined a markedly different cutoff for *PGR* (8.2 vs. 0.98, respectively), casting doubt on binary classification in this case being an optimal fit.

When analyzing the histograms of the mRNA expression data, we noted an overlap between normal and cancer datasets, as well as an asymmetry that signaled divergence in cancer (Fig. 2A–C). Further statistical testing clearly demonstrated that expression

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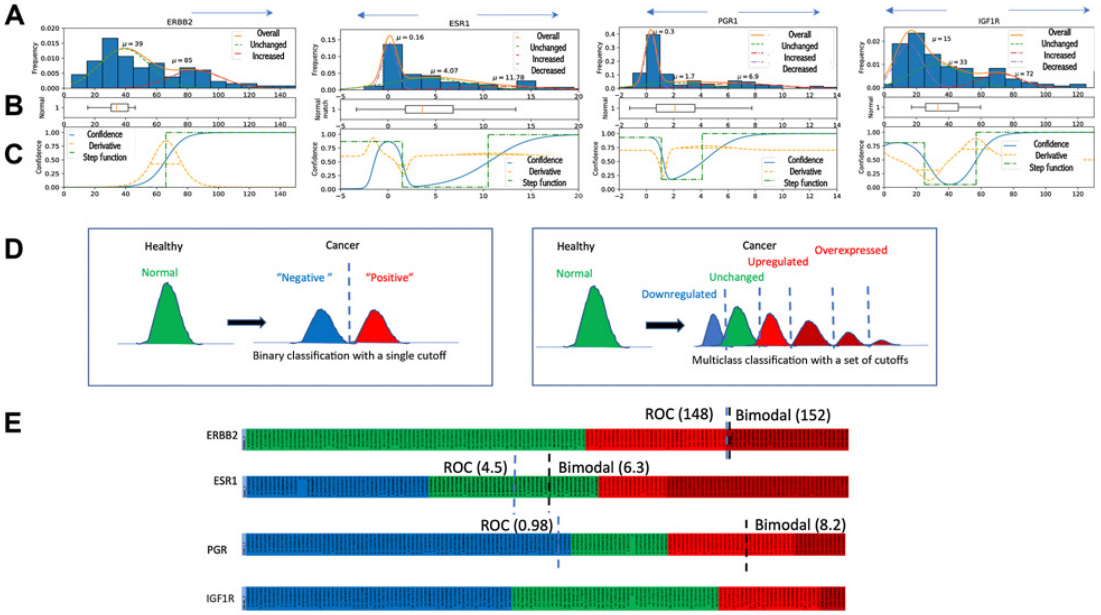

Figure 2.

GMM for *ERBB2, ESR1, PGR*, and *IGF1R* mRNA data. **A,** Gaussians for overall (orange), downregulated (green), and first upregulated (red) gene expression in cancer are superimposed on histograms of frequency of single-gene expression data. The adjusted R^2 for optimized fitting of the *ERBB2, ESR1, PGR*, and *IGF1R* histograms are 97.69%, 99.14%, 96.18%, and 92.62% respectively. **B,** Boxplots depict control measurements in mammoplasties for *ERBB2, PGR*, and *IGF1R* and in histologically normal matches for *ESR1* to reflect its dependence on age. Vertical lines in boxes, medians. Boxes, 25th and 75th percentile. Whiskers, 75th + 1.5xIQR and 25th − 1.5xIQR. *X* axis in **A** and **B,** mRNA levels in relative units. **C,** Graphs for confidence calculated for groups with changed gene expression as defined in Materials and Methods (blue). Step function plotted to illustrate minimal misclassification cutoffs as intersections of confidence and step function (green). The derivative of confidence with a half-height width plotted to define a range where the probability for sample to belong to either normal or changed group is similar (orange). **D,** Schemes of current and proposed models for cutoff determination in a continuous variable. The bimodal model, normal gene distribution in healthy tissue (green), "normal" (green), and "abnormal" (red) in cancer. The GMM, normal gene distribution in healthy tissue (green), downregulated (blue), unchanged (green), upregulated (red), and overexpressed (dark red) in cancer. **E,** Cancer classification based on minimal misclassification cutoffs for gene expression data: downregulated (blue), unchanged (green), upregulated (red), and overexpressed (dark red). Vertical dashed lines, cutoff computed through ROC model (blue) and bimodal model (black). The extrapolating of a single cutoff 152 (ROC model) or 148 (bimodal model) to GMM classifies cancer overexpressing *ERBB2* as $HER2^+$ and cancers with unchanged and upregulated levels of *ERBB2* as HER2. The dichotomization of the ESR1 data with either 4.5 (ROC model) or 6.3 (bimodal model) cutoff splits data in two groups, each of which contains either "false positive" or "false negative" cases with unchanged levels of *ESR1*. On the basis of the ROC model cutoff of 8.2, PR^+ treatment group would be comprised mainly of cancer with overexpressed *PGR*, whereas PR^- group would include cancers with downregulated, similar-to-normal, and upregulated, increasing the rate of "false negative." A bimodal model cutoff of 0.98 splits *PGR* data in to PR^- group comprised of the subpopulation with downregulated *PGR* and PR^+ group that apart from upregulated and overexpressed PR cases would certainly increase a rate of "false positive."

data for each gene mRNA across cancer samples showed variable degrees of skewness and did not pass normality tests (Supplementary Table S3). These observations led us to hypothesize that in cancer, gene expression data may conform to multiple standard Gaussian distributions, with one of these similar to the distribution in the normal dataset (Fig. 2D). The proposed GMM assumes the existence of more than one cutoff and more than two subpopulations in a single-gene expression data. We posited that cancers with "upregulated" gene expression exhibited a Gaussian distribution (or multiple distributions) centered on the mean value higher than the mean in "unchanged" group, whereas cancers with "downregulated" gene expression had a distribution around the mean value lower than that in "unchanged" group. Next, the empirical univariate mixtures of multiple curve fitting were computed to explore the unique shape of each gene expression data (as described in Materials and Methods). Finally, the PDFs were constructed to optimize the fit between the data and the probabilistic model. Using the single-gene PDF, we examined whether there was an "unchanged" component in the GMM for each gene expression data (Fig. 2A and B). The overlapping area between distributions of data from the normal data set and the GMM component potentially representing "unchanged" cancer data were 0.96, 0.95, 0.89, and 0.92 for *ERBB2, ESR1, PGR*, and *IGF1R*, respectively. Because overlapping area equals one only if two Gaussian distributions are exactly the same, these results supported our hypothesis that some cancer gene expression data fell within the range of values characteristic of normal data.

Next, the proportions of the breast cancers with downregulated and upregulated gene expression levels were estimated as areas under the PDF (Materials and Methods). *ERBB2* was unchanged in 52% (mean = 39.14), upregulated in 25% (mean = 84.85), and further upregulated in a series of low frequency distributions, representing 23% of total cases. This interesting result was based on the proven existence of a subset of cancer with unchanged

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Table 3. Cutoff values determination based on GMM minimal misclassification and confidence

	Minimal misclassification cutoff		Classification with confidence				
Gene	Low	High	Down group	Normal-down group	Normal group	Normal-up group	Up group
ERBB2	N.A.	66.0	N.A.	N.A.	<56.0	56.0–76.0	>76
ESR1	1.5	10.5	<1.0	1.0–2.3	2.3–7.3	7.3–14.3	>14.3
PGR	1.1	4.1	<0.9	0.9–1.6	1.6–2.5	2.5–6.3	>6.3
IGF1R	25.0	57.0	<20.0	20.0–35.0	35–48	48.0–66.0	>66.0

Abbreviation: N.A., not applicable.

ERBB2 and the use of adjusted R^2 that takes into account both the degree of freedom and the fitting quality (Supplementary Fig. S2A–S2C). In sharp contrast, *ESR1* was unchanged in 35% (mean = 4.07), decreased in 26% (mean = 0.16), increased in 12% (mean = 11.78), and overexpressed in 27% of IBC. *PGR* was unchanged in 22% (mean = 1.95), but markedly decreased to virtually undetectable levels in 42% (mean = 0.31), and increased in 26% (mean = 6.74), with further increase in 10% of cancer. Similarly, the *IGF1R* gene expression was unchanged in 39% (mean = 32.80), decreased in 40% (mean = 15.35), and increased in 19% (mean = 72.72), with overexpression in 2% of cancers.

Finally, to develop the MCPC and class identification criteria, we made a classifier from graphed confidence for changes in each gene expression (Fig. 2C). Mathematically, the intersection of two distributions represents a minimal misclassification point (Table 3). Individual cancers were classified into respective cancer subclasses using minimal misclassification points and compared with cutoffs determined through ROC and bimodal models. GMM classification with a set of cutoffs provided more detailed information about aberration in each gene expression in cancer (Fig. 2E). In some clinical applications, in order not to under- or overestimate target gene expression, lower and higher cutoff points of a higher confidence may be utilized for cancer gene classification. The subjectively selected cutoff point for the desired confidence may be used as an equivalent for classification criterion. In this article, they were computed for all target genes through the differentiation of confidence function followed by the calculation of the half-height width to define the range, where the probability of gene expression in cancer being changed and unchanged remains similar (Table 3). This range can be thought of as a tolerable interval of the misclassification error rate (a trade-off between misclassification of one type for misclassification of the other type). Thus, GMM appeared to be a well-suited model for representing nonnormal distributions of a single cancer gene expression data.

Assessment of GMM classifier using HER2, ER, PR, and IGF1R protein data

Aiming to validate whether *ESR1*, *PGR*, and *ERBB2* mRNA data represent main clinical IBC subtypes stratified on the basis of HR and HER2 status, the samples were grouped into ER^+ luminal A/B (LumA and LumB), $HER2^+$, and triple-negative (TN) subgroups (Table 4) Using minimal misclassification cutoff points, we calculated the fractions of samples with unchanged and changed gene expression levels in each subgroup. As expected, *ESR1* was upregulated in all ER^+ luminal subgroups and was downregulated in majority of ER^- subgroups, such as $HER2^+$ (62.50%) and TN (73.81%). *PGR* was upregulated in about half of luminal cases, whereas it was downregulated in the large portions of PR^- subgroups, including $HER2^+$ (75%), and TN (85.71%). Also as expected, *ERBB2* was upregulated in LumB (90.48%) as well as in $HER2^+$ (100%). In accord with the GMM prediction, there were no cancers with downregulated *ERBB2*. The subpopulations with unchanged *ERBB2* were found mainly in LumA (53.52%) and TN (88.10%) breast cancer subgroups.

Next, we investigated which breast cancer subgroups had changes in the expression of *IGF1R* measured synchronously with other genes (Table 4). Nearly half of the LumA cases had unchanged levels of *IGF1R*, whereas the rest had either upregulated (35.21%) or downregulated (18.31%) levels of *IGF1R*. In LumB subgroup, nearly a half (43%) of cases also had unchanged *IGF1R*, whereas the majority of the second half of LumB cases (47.62%) showed downregulated *IGF1R*. Furthermore, 73.81% of TN and 100% of $HER2^+$ breast cancer had downregulated *IGF1R*. In the analyzed cohort, *IGF1R* was sporadically overexpressed in about 4% of all IBC cases. Given the heterogeneity of breast cancer and the fact that interpatient variability may affect the analysis, we performed pairwise subgroup comparisons of 77 IBC and 77 matched noncancerous ipsilateral breast tissue

Table 4. Percent of patients with variations in a single-gene expression levels in invasive and *in situ* cancer subgroups

		Gene			
Group	Variation	*ESR1*	*PGR*	*ERBB2*	*IGF1R*
LumA	DOWN	7.04%	30.99%	0.00%	18.31%
	UC	26.76%	18.31%	53.52%	46.48%
	UP	**66.20%**	**50.70%**	46.48%	35.21%
LumB	DOWN	9.52%	**57.14%**	0.00%	47.62%
	UC	33%	24%	10%	43%
	UP	**57.14%**	19.05%	**90.48%**	9.52%
$HER2^+$	DOWN	**62.50%**	**75.00%**	0.00%	**100.00%**
	UC	37.50%	12.50%	0.00%	0.00%
	UP	0.00%	12.50%	**100.00%**	0.00%
TN	DOWN	**73.81%**	**85.71%**	0.00%	**73.81%**
	UC	26.19%	11.90%	88.10%	19.05%
	UP	0.00%	2.38%	11.90%	7.14%
$HER2^+$ CIS	DOWN	27.78%	**50.00%**	0.00%	**66.67%**
	UC	33.33%	38.89%	16.67%	11.11%
	UP	38.89%	11.11%	**83.33%**	22.22%
ER^+ CIS	DOWN	1.92%	21.15%	0.00%	9.62%
	UC	17.31%	19.23%	40.38%	42.31%
	UP	**80.77%**	**59.62%**	**59.62%**	48.08%
ER^- CIS	DOWN	**60.00%**	**80.00%**	0.00%	**80.00%**
	UC	40.00%	20.00%	60.0%	20.00%
	UP	0.00%	0.00%	40.0%	0.00%

NOTE: Change in a single gene found in ≥50% of patients per group is shown in bold. Luminal A, ER-positive/PR-positive and PR-negative/HER2-negative (ER^+/PR^+ and $PR^-/HER2^-$); luminal B, ER-positive/PR-positive/HER2-overexpressing ($ER^+/PR^+/HER2^+$); HER2-positive ($HER2^+$), ER-negative/PR-negative/HER2-overexpressing ($ER^-/PR^-/HER2^+$); triple negative (TN), ER-negative, PR-negative, and HER2 not overexpressing HER2 ($ER^-/PR^-/HER2^-$). $HER2^+$ CIS, HER2-overexpressing/ER-positive/ER-negative/PR-positive and PR-negative; ER^+ CIS, ER-positive/PR-positive/ and PR-negative/HER2 not overexpressing; ER^- CIS, ER-negative, PR-negative, and HER2 not overexpressing HER2.

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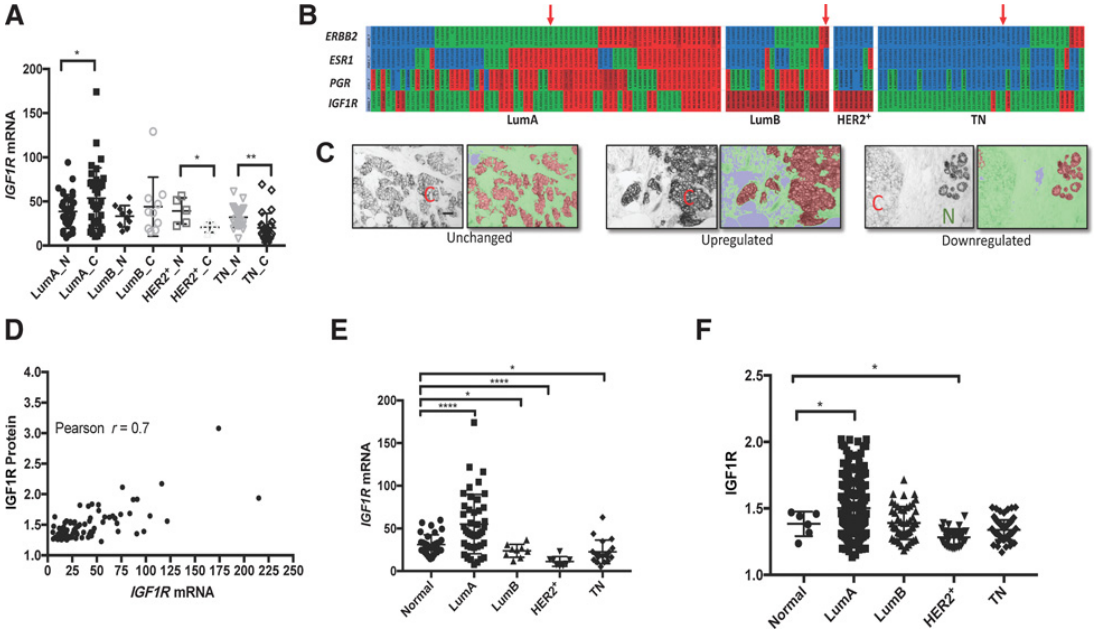

Figure 3.

Intrapatient and interpatient variations in *IGF1R* transcript and protein expression. **A,** Scatter plots show mean with SD for measurements of *IGF1R* mRNA in patient-matched groups of normal (N) and cancer (C) samples. *, $P < 0.05$; **, $P < 0.005$. **B,** Four-gene signatures for each cancer subgrouped as labeled. Color codes are as in Fig. 1E. Arrows indicate samples shown below. **C,** Representative original images (left) and overlays with the COMPAi-generated tissue segmentation maps used for quantification of IGF1R protein levels. C, areas of cancer; N, normal epithelia. Scale bar, 10 μm. Note: a loss of IGF1R expression in cancer compared with adjacent normal epithelium. **D,** Scatter dot plot shows a correlation between the *IGF1R* transcript and protein levels. *x*-axis, the relative expression of *IGF1R* transcript; *y*-axis, the relative expression of *IGF1R* protein in the same cancer samples ($n = 81$); *r*, the Pearson correlation coefficient between two data sets. Scatter plots depict mean with SD for measurements of the *IGF1R* transcript (**E**) and protein (**F**) in IBC subgroups. *, $P < 0.05$*; ****, $P < 0.0001$.

samples (Fig. 3A). The *IGF1R* mRNA expression levels were significantly upregulated in LumA, but downregulated in HER2$^+$ and TN cancers, as compared with noncancerous tissue from the same breast. A similar trend for *IGF1R* downregulation in conjunction with overexpression of *ERRB2* and loss of HR gene expression was observed in the independent cohort of preinvasive cancer (Table 4).

Because mRNA data do not always indicate protein expression, we aimed to examine whether the expression of *IGF1R* mRNA was associated with the expression of the IGF1R. Currently, there are no standard criteria for IGF1R positivity by IHC (22). To minimize subjectivity of the visual assessment of IHC staining and derive continuous readout for the relative expression of IGF1R, we utilized image-based quantification as described previously (26, 31). Moreover, because the results of QG2 assay do not distinguish between the gene expression in the tumor cells from that of the microenvironment, such as stromal and inflammatory cells, we performed measurements of IGF1R after digital mapping of the areas involved by cancer (Fig. 3B and C). IGF1R is known to localize to the cell membrane, cytoplasm, and nucleolus (32), thus our measurements reflected the aggregate protein expression in all the cellular compartments. Notably, IGF1R immunoreactivity was observed in cancer with unchanged and upregulated levels of *IGF1R* mRNA; however, it was lost in cases with downregulated *IGF1R* mRNA in stark contrast to adjacent normal epithelium where IGF1R was normally expressed (Fig. 3C). As illustrated in Fig. 3D, *IGF1R* mRNA measured by QG2 assay and the average value for IGF1R protein expression showed a strong linear correlation ($r = 0.68$; $P = 2.12204\text{E-}12$), thus allowing the IGF1R protein to be a second criterion in cancer classification. A comparison of the mean value in each breast cancer subgroup with those in the healthy tissue groups revealed that protein and mRNA levels of *IGF1R* were significantly upregulated in LumA, downregulated in TN, and significantly downregulated in HER2$^+$ subgroup of IBC (Fig. 3E and F). Thus, the results of inter- and intrapatient analyses clearly demonstrated that in less aggressive breast cancer, IGF1R levels were either similar-to-normal or upregulated (rarely overexpressed), whereas in more aggressive breast cancer IGF1R level is frequently below the levels found in the healthy breast epithelia.

Discussion

Differences between the GMM classifier and current classification methods

This article presents the original probabilistic GMM-based multiclass classifier aimed to extract biologically relevant and potentially clinically useful information from molecular datasets obtained from both healthy and cancerous tissue. Although there is a range of statistical methods for prognostic cutoff point selection, including optimization of *P* value (33) and maximization of combination of test sensitivity and specificity (34), most of

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them produce binary (27) and some tripartite (35) divisions of data, based on the correlation with clinical outcome or response to treatment. The proposed classifier is fundamentally different from the published cutoff finders. It is also different from a Bayesian GMM-based clustering method that aims to subgroup patients based on similarity in patterns of expression among multiple genes (36). The proposed finite GMM method uses probability theory to make inferences about a number of classes in cancer, without predefined knowledge of that number. Specifically, the GMM classifier allows explicit analysis of mRNA profiles, each of which is comprised of the expression values for a single gene across the entire range of samples. It performs empirical curve fitting and identifies distributions that most accurately model the distributions of the actual expression data. It calculates multiple cutoffs to allow classification of individual cancer into the respective subclass. Furthermore, this classifier outputs the value of confidence to estimate confidence in the cutoff prediction, which is a measure of predictive accuracy. Altogether, these fundamental differences make the GMM classifier effective tool for clinical prediction.

The comparison of results from GMM and binary models

In this article, the GMM classifier identified breast cancer subclasses in which the abundance of transcripts encoding HER2 were unchanged and upregulated, whereas the abundance of messengers encoding HR were unchanged and dually dysregulated, shifting either up or down relative to those in healthy breast tissue. These findings concur with the results from genomic and FISH studies that have documented the amplification of *ERBB2* (37) and complex dysregulations in HR genes due to variations in copy number (deletion, neutral, gain, and amplification) in correlation with mRNA levels (38, 39). Because proteins, rather than transcripts, are ultimate diagnostic, prognostic, and treatment response predictors, and considerable efforts have been dedicated to validating the RNA-based approach for assessment of receptor status (6, 7). High, moderate, and low agreement with IHC has been shown for *ERBB2*, *ESR1*, and *PGR*, respectively, regardless of the any given study cohort and technique, including DNA microarray (7) and PCR-based assays such as OncotypeDX (40) and Xpert Breast Cancer STRAT4 (41). The results of direct comparison of the GMM and binary classifier performances in this article suggest that discrepant findings in early reports are likely due to analytical rather than technical reasons. The magnitude of *ERBB2* gene amplification often results in marked overexpression of HER2 (37). Consequently, all three models (ROC, bimodal, and GMM) fitted to *ERBB2* mRNA data determined *ERBB2*–overexpressing cancers that were also $HER2^+$ on pathology reports. The dichotomization of structurally mixed data of *ESR1* and *PGR* resulted in the loss of reliable information about patient groups, as well as misclassification of some individuals. Indeed, the best ROC cutoff for *ESR1* data divided cases with unchanged *ESR1* mRNA, thus "contaminating" ER^+ and ER^- groups. Moreover, with ER positivity scoring criterion (10% or greater nuclear staining for ER) used by many clinicians to consider patient eligibility for endocrine therapy (42), the ER^+ group was represented by tumors with unchanged, upregulated, and overexpressed *ESR1* that may plausibly constitute distinctive groups with respect to ER content and sensitivity to hormone therapy. Likewise, the GMM classifier distinguished a single PR^- group ($\leq$1% nuclear staining for PR by IHC), but parsed the PR^+ group into dissimilar subgroups containing cases with unchanged, upregulated, and overexpressed *PGR*. Thus, GMM-based classification of primary breast cancer is consistent with traditional binary classification on standard clinical markers such as HR and HER2, yet it provides an additional layer of information relying on the probability estimates of all events.

Novel findings revealed by the GMM classifier and their significance

The most striking finding revealed by the GMM classifier is the discovery of a fundamental difference in cancer-related variations of *ERBB2* and *IGF1R* genes encoding HER2 and IGF1R. Both transmembrane receptors have important normal functions in mammary gland morphogenesis, and their dysregulation may be detrimental for the glandular structure and function of the breast (43, 44). For years, both the HER2 and IGF1R have been studied under the paradigm of receptor overexpression contributing to cancer progression through receptor-mediated signaling (45). Despite great appeal of targeting IGF1R in breast cancer, discrepant data have been published on a prognostic significance of IGF1R (22), and the clinical trials testing anti-IGF1R therapies were remarkably unsuccessful (23). Of note, overexpressed HER2 was found consistently in about 20% of all IBC (37), whereas the reported prevalence of "high" IGF1R ranged from 39% to 93%, suggesting that determination of IGF1R was hampered by technical and interpreter issues (46). As estimated by the GMM classifier, among all the human IBC, about 40% are likely to have normal, 40% downregulated, and about 20% higher than normal levels of *IGF1R*. The existence of *IGF1R* "down" subclass of breast cancer is strongly supported by inter- and intrapatient analyses and has several implications. First, low *IGF1R* is a common feature of the most aggressive IBC, including but not limited to $HER2^+$, TN, and Lum B cancer subtypes. These findings are in accord with early report on inverse association of IGF1R expression with proliferative activity of more aggressive breast tumors (47). Second, lower *IGF1R* associates with higher histologic grade, a clinical marker of cancer dedifferentiation and aggressiveness in IBC (48). This suggests that IGF1R is a marker of differentiation, and that its loss or underexpression may be involved in the pathogenesis of the subset of IBC. And third, the observed low *IGF1R* in carcinoma *in situ* suggests that loss of this receptor expression may be a very early alteration in some breast cancers. Our results from diagnostic preoperative biopsies are particularly clinically important, as patients with poorly differentiated (high nuclear grade) of ductal carcinoma *in situ* (DCIS) have an increased risk of ipsilateral breast tumor recurrence compared with patients diagnosed with well-differentiated (low grade) DCIS and may be treated differently (49). It is worth noting that although the upregulation of *IGF1R* in human IBC may occur less frequently than has been thought, the rate of *IGF1R* overexpression forecasted by the GMM classifier is similar to the previously reported sporadic *IGF1R* amplification associated with the IGF1R protein overexpression and a shorter median overall survival (50). Likewise, The Cancer Genome Atlas and the Catalog of Somatic Mutations in Cancer reported the low frequency of *IGF1R* amplification (2%–5%) relying on the observational data from thousands of tissue samples. Importantly, our study provides a cost-effective method for calculating MCPC and thereby pinpoints the patient cohort with abnormally high levels of IGF1R in their tumors and provides a mean for exclusion of patients with normal or lower than normal levels of this target in their tumors. Accurate characterization of tumor is critical not only for better

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understanding cancer biology, but also for targeted enrollment into clinical trials and for developing personalized therapies to improve patient outcome.

In sum, the presented GMM-based classifier considers the common problem of the biological cutoff and class discovery in cancer and provides a "first aid" solution for resolving ambiguity of a candidate tumor marker. Because of the general flexibility of GMM, this classifier could be applicable to a large number of gene expression data currently awaiting assessments of clinical value.

Disclosure of Potential Conflicts of Interest

No potential conflicts of interest were disclosed.

Authors' Contributions

Conception and design: M.A. Guvakova
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Acquisition of data (provided animals, acquired and managed patients, provided facilities, etc.): I. Prabakaran, B. Tong, G. Koo, P.J. Zhang
Analysis and interpretation of data (e.g., statistical analysis, biostatistics, computational analysis): I. Prabakaran, Z. Wu, C. Lee, S. Steeman, M.A. Guvakova
Writing, review, and/or revision of the manuscript: I. Prabakaran, Z. Wu, B. Tong, S. Steeman, G. Koo, M.A. Guvakova
Administrative, technical, or material support (i.e., reporting or organizing data, constructing databases): I. Prabakaran, Z. Wu
Study supervision: M.A. Guvakova

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